

1.



SIMATS
ENGINEERING

SIMATS Online Programming Lab
C++ PROGRAMMING

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QUESTIONS 19 / 19 completed

Plus

PLUS

Write a C++ program to overload the + operator to add two complex numbers.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 class Complex {
5     float real;
6     float imag;
7
8 public:
9     Complex(float r = 0, float i = 0) {
10         real = r;
11         imag = i;
12     }
13
14     // Overloading + operator
15     Complex operator+(const Complex& obj) {
16         Complex temp;
17         temp.real = real + obj.real;
18         temp.imag = imag + obj.imag;
19         return temp;
20     }
21 }
```

INPUT

3.4 2.1
1.2 4.5

2.



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Increment

INCREMENT

Write a C++ program to overload the ++ operator to increment a variable.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 class Number {
5     int value;
6
7 public:
8     Number(int v = 0) {
9         value = v;
10    }
11
12    // Overloading prefix ++ operator
13    Number operator++() {
14        value++;
15        return *this;
16    }
17
18    void display() {
19        cout << value << endl;
20    }
21 };
```

INPUT

5

3.

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<< >>

<<

Write a C++ program to overload the << operator to print the contents of a user-defined class.

PROGRAM EDITOR

C++

Run Save

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 class Person {
6     string name;
7     int age;
8
9 public:
10    Person(string n = "", int a = 0) {
11        name = n;
12        age = a;
13    }
14
15    // Overloading << operator as a friend function
16    friend ostream& operator<<(ostream& out, const Person& p) {
17        out << "Name: " << p.name << endl;
18        out << "Age: " << p.age;
19        return out;
20    }
21 }
```

INPUT

Alice 21

4.



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QUESTIONS 19 / 19 completed

Equal

EQUAL

Write a C++ program to overload the == operator to compare two objects of a user-defined class.

PROGRAM EDITOR

C++

Run Save

```

1 #include <iostream>
2 using namespace std;
3
4 class Number {
5     int val;
6
7 public:
8     Number(int v = 0) {
9         val = v;
10    }
11
12    // Overloading == operator
13    bool operator==(const Number& obj) {
14        return val == obj.val;
15    }
16 };
17
18 int main() {
19     int a, b;
20     cin >> a >> b;
21

```

INPUT

10 10

5.



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QUESTIONS 19 / 19 completed

[]

[]

Write a C++ program to overload the [] operator to access the elements of an array using an index.

PROGRAM EDITOR

C++

Run Save

```

1 #include <iostream>
2 using namespace std;
3
4 class Number {
5     int val;
6
7 public:
8     Number(int v = 0) {
9         val = v;
10    }
11
12    // Overloading == operator
13    bool operator==(const Number& obj) {
14        return val == obj.val;
15    }
16 };
17
18 int main() {
19     int a, b;
20     cin >> a >> b;
21

```

INPUT

10 10

6.



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QUESTIONS 19 / 19 completed

*

*

Write a C++ program to overload the * operator to multiply two matrices.

PROGRAM EDITOR

C++

Run

Save

INPUT

2 2
1 2
3 4
2 2
5 6
7 8